◆ AWS CLOUD PRACTITIONER BOOTCAMP

WEEK 3 — NETWORKING + ARCHITECTURE THINKING

SESSION 7 — NETWORKING FUNDAMENTALS

SESSION NOTES

◆ SESSION OBJECTIVE

This session introduces the foundations of AWS networking and cloud architecture thinking.

Students must understand:

✓ What networking means in cloud computing
✓ Why Amazon VPC exists
✓ The purpose of subnets
✓ Difference between public and private subnets
✓ How Internet Gateways work
✓ Why networking affects security

◆ WHAT IS NETWORKING?

Networking is the system that allows devices and cloud resources to communicate.

In AWS, networking controls:

◆ connectivity
◆ communication
◆ security
◆ traffic flow
◆ internet access

Without networking:

- applications cannot communicate
- users cannot access services
- cloud systems become isolated

◆ WHY NETWORKING MATTERS

Networking is one of the most important parts of cloud architecture because it controls:

✓ who can access systems
✓ what remains private
✓ how traffic flows
✓ how applications communicate
✓ how attacks are minimized

Poor networking design can lead to:

✕ exposed databases
✕ unauthorized access
✕ downtime
✕ security breaches

◆ AMAZON VPC (VIRTUAL PRIVATE CLOUD)

Definition

Amazon VPC allows organizations to create isolated private networks inside AWS.

Think of a VPC as:

Your private data center inside the AWS cloud.

Inside a VPC, organizations deploy:

✓ EC2 instances
✓ databases
✓ applications
✓ load balancers

◆ WHY VPC IS IMPORTANT

Amazon VPC provides:

✓ isolation
✓ security
✓ infrastructure control
✓ traffic management

Without VPCs:

- resources would not be logically separated

- security boundaries would become weak
- infrastructure management would become difficult

◆ VPC COMPONENTS

A VPC commonly contains:

Component	Purpose
Subnets	Segment the network
Route Tables	Control traffic routing
Internet Gateway	Enable internet connectivity
Security Groups	Instance-level firewall
NACLs	Subnet-level security
EC2 Instances	Compute resources

◆ WHAT IS A SUBNET?

A subnet is:

A smaller network inside a VPC.

Subnets help organizations:

✓ organize infrastructure
✓ separate workloads
✓ improve security
✓ manage traffic flow

◆ PUBLIC SUBNET

A public subnet allows resources to communicate with the internet.

Examples:
✓ web servers
✓ load balancers

Characteristics:
✓ internet accessible
✓ public IP addresses possible
✓ external traffic allowed

◆ PRIVATE SUBNET

A private subnet hides resources from direct internet access.

Examples:
✓ databases
✓ internal applications
✓ backend systems

Characteristics:
✓ more secure
✓ no direct internet exposure
✓ sensitive workloads protected

◆ PUBLIC VS PRIVATE SUBNETS

Public Subnet	Private Subnet
Internet accessible	No direct internet access
Hosts web applications	Hosts databases
Higher exposure	More secure
Public-facing services	Internal systems

◆ INTERNET GATEWAY (IGW)

An Internet Gateway allows communication between a VPC and the internet.

Without an IGW:
✕ users cannot access applications
✕ EC2 instances cannot access the internet

◆ HOW TRAFFIC FLOWS

Example AWS traffic flow:

```
User → Internet → Internet Gateway → Public Subnet → EC2
```

This process allows users to access applications hosted in AWS.

◆ REAL-WORLD EXAMPLE

Online Banking Platform

Public Subnet

Hosts:
✓ web applications
✓ customer login portal

Private Subnet

Hosts:
✓ databases
✓ payment systems
✓ internal APIs

Why?

Sensitive systems should not be publicly exposed.

◆ NETWORKING SECURITY RISKS

Common networking mistakes include:

✕ public database exposure
✕ open SSH ports
✕ unrestricted internet access
✕ poor subnet segmentation

These mistakes can cause:
✓ data breaches
✓ unauthorized access
✓ service compromise

◆ NETWORKING BEST PRACTICES

✓ Use private subnets for sensitive systems
✓ Limit internet exposure
✓ Separate workloads logically
✓ Apply least privilege principles
✓ Monitor network traffic
✓ Document architecture clearly

◆ GOVERNANCE INSIGHT

Networking is not only about connectivity.

Networking controls:

✓ security
✓ isolation
✓ risk exposure
✓ system communication
✓ architecture quality

Good networking design creates secure cloud foundations.

◆ KEY TAKEAWAYS

✓ Amazon VPC creates isolated cloud networks
✓ Subnets organize infrastructure securely
✓ Public subnets allow internet access
✓ Private subnets protect sensitive systems
✓ Internet Gateways enable connectivity
✓ Networking directly affects cloud security

◆ FINAL INSIGHT

“If you cannot design the network correctly, you cannot secure the system correctly.”

Cloud architecture begins with networking foundations.